Homework 4

BME 5910 - Biomedical Optics

Garrett Kunkler

9/28/2026

Problem 1

```
% Givens
mu_x = 0.3;
mu_y = 0.4;
mu_z = 0.866;
theta = 30;
psi = 60;

% Solution
mu_x_prime = sind(theta)*cosd(psi)
```

```
mu_x_prime =
0.2500
```

```
mu_y_prime = sind(theta)*sind(psi)
```

```
mu_y_prime =
0.4330
```

```
mu_z_prime = cosd(theta)
```

```
mu_z_prime =
0.8660
```

```
s_prime = [mu_x_prime; mu_y_prime; mu_z_prime]
```

```
s_prime = 3×1
0.2500
0.4330
0.8660
```

```

theta_0 = acosd(mu_z)
```

```
theta_0 =
30.0029
```

```
phi_0 = acosd(0.3/sind(theta_0))
```

```
phi_0 =
53.1339
```

```

R_y = [cosd(theta_0)  0 sind(theta_0);
       0              1 0;
       -sind(theta_0) 0 cosd(theta_0)]
```

```
R_y = 3×3
0.8660     0          0.5000
0          1.0000     0
-0.5000    0          0.8660
```

```
R_z = [cosd(phi_0) -sind(phi_0) 0;
       sind(phi_0) cosd(phi_0)  0;
       0           0            1]
```

```
R_z = 3×3
0.5999     -0.8000    0
0.8000     0.5999     0
0          0          1.0000
```

```
T_inv = R_z*R_y
```

```
T_inv = 3×3
0.5196     -0.8000    0.3000
0.6928     0.5999     0.4001
-0.5000    0          0.8660
```

```
s_new = T_inv*s_prime
```

```
s_new = 3×1
0.0433
0.7795
0.6250
```

```
mu_x_new = s_new(1)
```

```
mu_x_new =
0.0433
```

```
mu_y_new =  s_new(2)
```

```
mu_y_new =
0.7795
```

```
mu_z_new =  s_new(3)
```

```
mu_z_new =
0.6250
```

```
s_new_mag_squared = mu_x_new^2 + mu_y_new^2 + mu_z_new^2
```

```
s_new_mag_squared =
1.0000
```

Problem 2

```
% Givens
n_1 = 1.37;
```

```
mu_a_1 = 0.02;
mu_s_1 = 10;
g_1 = 0.9;

z_b = 1;

n_2 = 1.45
```

```
n_2 =
1.4500
```

```
mu_a_2 = 0.04;
mu_s_2 = 15;
g_2 = 0.85
```

```
g_2 =
0.8500
```

```
k_start = [0.2; 0.1; 0.92];
s_start = [0.3; 0.4; 0.866];

xi = 0.3;
xi_R = 0.02;

% Solution
mu_t_1 = mu_a_1 + mu_s_1
```

```
mu_t_1 =
10.0200
```

```
mu_t_2 = mu_a_2 + mu_s_2
```

```
mu_t_2 =
15.0400
```

```
s = -log(xi)/mu_t_1
```

```
s =
0.1202
```

```
s_b = (z_b - k_start(3))/s_start(3)
```

```
s_b =
0.0924
```

```
k_b = k_start + s_b * s_start
```

```
k_b = 3×1
    0.2277
    0.1370
    1.0000
```

```
theta_i = acosd(s_start(3))
```

```
theta_i =
30.0029
```

```
phi_i = acosd(s_start(1)/sind(theta_i))
```

```
phi_i =
53.1339
```

```
theta_t = asind(n_1/n_2*sind(theta_i))
```

```
theta_t =
28.1938
```

```
R_s = ((n_1*cosd(theta_i) - n_2*cosd(theta_t))/(n_1*cosd(theta_i) +
n_2*cosd(theta_t)))^2
```

```
R_s =
0.0014
```

```
R_p = ((n_1*cosd(theta_t) - n_2*cosd(theta_i))/(n_1*cosd(theta_t) +
n_2*cosd(theta_i)))^2
```

```
R_p =
3.8362e-04
```

```
R = (R_s + R_p)/2
```

```
R =
8.8176e-04
```

```
s_new = [sind(theta_t)*cosd(phi_i);
         sind(theta_t)*sind(phi_i);
         cosd(theta_t)]
```

```
s_new = 3×1
    0.2834
    0.3780
    0.8814
```

```
s_sb = mu_t_1/mu_t_2*(s-s_b)
```

```
s_sb =
0.0185
```

```
k_end = k_b + s_sb*s_new
```

```
k_end = 3×1
    0.2330
    0.1439
    1.0163
```